

MCU (ESP32-S3): Working Document

Digital Audio Player • Rev 3

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary	1
2. Design Requirements	1
3. Pin Configuration	2
3.1 IO Pin Default Configuration	2
3.2 Power-Up Glitches on Pins	4
3.3 GPIO Assignment	4
4. External Components	6
5. Equations and Results	7
6. Guideline Notes	7
7. Open Problems	7
8. Revision Log	8
9. Sources	8

1. Summary

Field	Value
Manufacturer / MPN	Espressif Systems, ESP32-S3 (base chip)
Package	56-QFN, 7 × 7 mm, exposed pad (pin 57, GND)
Memory	No in-package flash or PSRAM; external SPI flash required
Function	Main controller: file decode, I2S to the DAC, UI, charger control, USB mass storage
Clock	External 40 MHz passive crystal (compulsory), ECS-400-10-37B2-CKY-TR, $C_L = 10$ pF

Source: [2], [3], [4]

2. Design Requirements

Requirement	Value	Source
Supply	+3V3_DIG; 3.0 to 3.6 V on all supply pins	[2]
Crystal	40 MHz only; accuracy within ± 10 ppm	[2]
Storage	microSD over 4-bit SDMMC	Author

Requirement	Value	Source
Audio	I2S slave to the ES9038Q2M (DAC is master)	Author
Control	I2C to the DAC and the BQ24250 at 100 kHz	Author
Display	4-wire SPI to the ER-TFT024IPS-3	Author
USB	USB-C, mass storage (GPIO19 / GPIO20)	Author, [2]
Constraints	Low noise · Small footprint	Author

3. Pin Configuration

3.1 IO Pin Default Configuration

No.	Name	Power	At Reset	After Reset
1	LNA_IN			
2	VDD3P3			
3	VDD3P3			
4	CHIP_PU	VDD3P3_RTC		
5	GPIO0	VDD3P3_RTC	IE, WPU	IE, WPU
6	GPIO1	VDD3P3_RTC	IE	IE
7	GPIO2	VDD3P3_RTC	IE	IE
8	GPIO3	VDD3P3_RTC	IE	IE
9	GPIO4	VDD3P3_RTC		
10	GPIO5	VDD3P3_RTC		
11	GPIO6	VDD3P3_RTC		
12	GPIO7	VDD3P3_RTC		
13	GPIO8	VDD3P3_RTC		
14	GPIO9	VDD3P3_RTC		IE
15	GPIO10	VDD3P3_RTC		IE
16	GPIO11	VDD3P3_RTC		IE
17	GPIO12	VDD3P3_RTC		IE
18	GPIO13	VDD3P3_RTC		IE
19	GPIO14	VDD3P3_RTC		IE
20	VDD3P3_RTC			
21	XTAL_32K_P	VDD3P3_RTC		
22	XTAL_32K_N	VDD3P3_RTC		
23	GPIO17	VDD3P3_RTC		IE
24	GPIO18	VDD3P3_RTC		IE
25	GPIO19	VDD3P3_RTC		
26	GPIO20	VDD3P3_RTC	USB_PU	USB_PU
27	GPIO21	VDD3P3_RTC		
28	SPICS1	VDD_SPI	IE, WPU	IE, WPU

No.	Name	Power	At Reset	After Reset
29	VDD_SPI			
30	SPIHD	VDD_SPI	IE, WPU	IE, WPU
31	SPIWP	VDD_SPI	IE, WPU	IE, WPU
32	SPICS0	VDD_SPI	IE, WPU	IE, WPU
33	SPICLK	VDD_SPI	IE, WPU	IE, WPU
34	SPIQ	VDD_SPI	IE, WPU	IE, WPU
35	SPID	VDD_SPI	IE, WPU	IE, WPU
36	SPICLK_N	VDD_SPI / VDD3P3_CPU	IE	IE
37	SPICLK_P	VDD_SPI / VDD3P3_CPU	IE	IE
38	GPIO33	VDD_SPI / VDD3P3_CPU		IE
39	GPIO34	VDD_SPI / VDD3P3_CPU		IE
40	GPIO35	VDD_SPI / VDD3P3_CPU		IE
41	GPIO36	VDD_SPI / VDD3P3_CPU		IE
42	GPIO37	VDD_SPI / VDD3P3_CPU		IE
43	GPIO38	VDD3P3_CPU		IE
44	MTCK	VDD3P3_CPU		IE
45	MTDO	VDD3P3_CPU		IE
46	VDD3P3_CPU			
47	MTDI	VDD3P3_CPU		IE
48	MTMS	VDD3P3_CPU		IE
49	U0TXD	VDD3P3_CPU	IE, WPU	IE, WPU
50	U0RXD	VDD3P3_CPU	IE, WPU	IE, WPU
51	GPIO45	VDD3P3_CPU	IE, WPD	IE, WPD
52	GPIO46	VDD3P3_CPU	IE, WPD	IE, WPD
53	XTAL_N			
54	XTAL_P			
55	VDDA			
56	VDDA			
57	GND			

IE: input enabled. WPU: internal weak pull-up enabled. WPD: internal weak pull-down enabled. USB_PU: USB pull-up resistor enabled.
Source: [2], section GPIO.

3.2 Power-Up Glitches on Pins

Pin	Glitch ^a	Typical Time (μs)
GPIO1	Low-level glitch	60
GPIO2	Low-level glitch	60
GPIO3	Low-level glitch	60
GPIO4	Low-level glitch	60
GPIO5	Low-level glitch	60
GPIO6	Low-level glitch	60
GPIO7	Low-level glitch	60
GPIO8	Low-level glitch	60
GPIO9	Low-level glitch	60
GPIO10	Low-level glitch	60
GPIO11	Low-level glitch	60
GPIO12	Low-level glitch	60
GPIO13	Low-level glitch	60
GPIO14	Low-level glitch	60
XTAL_32K_P	Low-level glitch	60
XTAL_32K_N	Low-level glitch	60
GPIO17	Low-level glitch	60
GPIO18	Low-level / High-level glitch	60
GPIO19	Low-level glitch / High-level glitch ^b	60
GPIO20	Pull-down glitch / High-level glitch ^b	60

^a Low-level: pin outputs low for the period. High-level: pin outputs high for the period. Pull-down: pin is held by the internal weak pull-down for the period. ^b GPIO19 and GPIO20 each show two high-level glitches of about 60 μs during power-up; total duration and delay are 3.2 ms (GPIO19) and 2 ms (GPIO20).

Source: [2], section GPIO, table Power-Up Glitches on Pins.

3.3 GPIO Assignment

Current connection of every GPIO-capable pin, read from the schematic sheets. **X** marks a recommended change; the comment ID points to the table that follows. Reset state is at reset / after reset, from Section 3.1 (Hi-Z: no input, no pull). Sheets: MCU-3, Audio-1, Power-2, Periph-4.

Pin	Name	Net / connection	Sheet	Dir	Reset	External pull	Flags	Sw.	Cmt.
5	GPIO0	NC			IE, WPU	None	Strap (boot)	X	C1
6	GPIO1	GPIO1_DAC	Audio-1	I/O	IE	10 kΩ pd (R13)	Glitch		C2, C6
7	GPIO2	GPIO2_DAC	Audio-1	I/O	IE	10 kΩ pd (R14)	Glitch		C2, C6
8	GPIO3	SELECT	Periph-4	In	IE	10 kΩ pd	Strap (JTAG); glitch; RTC		C3, C12
9	GPIO4	SDA_DAC / SDA_BQ	Audio-1, Power-2	I/O	Hi-Z	10 kΩ pu × 2	Glitch		C4
10	GPIO5	SCL_DAC / SCL_BQ	Audio-1, Power-2	Out	Hi-Z	10 kΩ pu × 2	Glitch		C4
11	GPIO6	ADDR_DAC	Audio-1	Out	Hi-Z	10 kΩ pu (R9)	Glitch		C6
12	GPIO7	DATA_CLK_DAC (I2S bit clock)	Audio-1	In	Hi-Z	None	Glitch		
13	GPIO8	DATA1_DAC (I2S frame clock)	Audio-1	In	Hi-Z	None	Glitch		

Pin	Name	Net / connection	Sheet	Dir	Reset	External pull	Flags	Sw.	Cmt.
14	GPIO9	DATA2_DAC (I2S data)	Audio-1	Out	Hi-Z / IE	None	Glitch		
15	GPIO10	PG_+3V3	Power-2	In	Hi-Z / IE	10 kΩ pu (R224)	Glitch		
16	GPIO11	PG_-3V3	Power-2	In	Hi-Z / IE	10 kΩ pu (R222)	Glitch		
17	GPIO12	TRI_STATE_24	Audio-1	Out	Hi-Z / IE	10 kΩ pd (R8)	Glitch		
18	GPIO13	TRI_STATE_22	Audio-1	Out	Hi-Z / IE	10 kΩ pd (R7)	Glitch		
19	GPIO14	VOLUME+	Periph-4	In	Hi-Z / IE	10 kΩ pd	Glitch; RTC		C12
21	GPIO15 (XTAL_32K_P)	NC; proposed ENC_A		In	Hi-Z	Per encoder	Glitch; RTC	X	C5
22	GPIO16 (XTAL_32K_N)	NC; proposed ENC_B		In	Hi-Z	Per encoder	Glitch; RTC	X	C5
23	GPIO17	VOLUME-	Periph-4	In	Hi-Z / IE	10 kΩ pd	Glitch; RTC		C12
24	GPIO18	RESET_LCD	Periph-4	Out	Hi-Z / IE	None	Glitch (low / high)		C8
25	GPIO19	DN (USB D-)	Power-2	I/O	Hi-Z	22 Ω series	Glitch; USB		
26	GPIO20	DP (USB D+)	Power-2	I/O	USB_PU	22 Ω series	Glitch; USB		
27	GPIO21	POWER	Periph-4	In	Hi-Z	10 kΩ pd	RTC (wake)		C12
28	SPICS1 (GPIO26)	NC; proposed SD_CD		In	IE, WPU	None	PSRAM CS if fitted	X	C6
30	SPIHD (GPIO27)	NC; flash HOLD / IO3		I/O	IE, WPU		Flash	X	C9
31	SPIWP (GPIO28)	NC; flash WP / IO2		I/O	IE, WPU		Flash	X	C9
32	SPICS0 (GPIO29)	Wire stub; flash CS	MCU-3	Out	IE, WPU		Flash	X	C9
33	SPICLK (GPIO30)	Wire stub; flash CLK	MCU-3	Out	IE, WPU		Flash	X	C9
34	SPIQ (GPIO31)	Wire stub; flash DO / IO1	MCU-3	I/O	IE, WPU		Flash	X	C9
35	SPID (GPIO32)	NC; flash DI / IO0		I/O	IE, WPU		Flash	X	C9
36	SPICLK_N (GPIO48)	NC; proposed BL_PWM		Out	IE	None	VDD_SPI domain	X	C7
37	SPICLK_P (GPIO47)	NC; proposed SD_CLK		Out	IE	None	VDD_SPI domain	X	C6
38	GPIO33	PLUG_DET	Audio-1	In	Hi-Z / IE	100 kΩ pu (R19), 10 kΩ series (R20)			
39	GPIO34	EN_SPEAKER	Audio-1	Out	Hi-Z / IE	50 kΩ pd (R31) via 54.9 kΩ (R29)			
40	GPIO35	SHDN_BB_3.9	Power-2	Out	Hi-Z / IE	None		X	C10
41	GPIO36	RESETB_DAC	Audio-1	Out	Hi-Z / IE	10 kΩ pd (R12)			C2
42	GPIO37	INT_BQ	Power-2	In	Hi-Z / IE	10 kΩ pu			
43	GPIO38	D/CX_LCD	Periph-4	Out	Hi-Z / IE	None			
44	MTCK (GPIO39)	SCL_LCD	Periph-4	Out	Hi-Z / IE	None	JTAG pin		
45	MTDO (GPIO40)	MOSI_LCD	Periph-4	Out	Hi-Z / IE	None	JTAG pin		
47	MTDI (GPIO41)	NC; proposed SD_CMD		I/O	Hi-Z / IE	Pull-up	JTAG pin	X	C6
48	MTMS (GPIO42)	NC; proposed SD_D0		I/O	Hi-Z / IE	Pull-up	JTAG pin	X	C6
49	U0TXD (GPIO43)	DOWN	Periph-4	In	IE, WPU	10 kΩ pd	UART0 TX	X	C11, C12
50	U0RXD (GPIO44)	UP	Periph-4	In	IE, WPU	10 kΩ pd	UART0 RX		C11, C12
51	GPIO45	CS_LCD	Periph-4	Out	IE, WPD	10 kΩ pd	Strap (VDD_SPI)		C13
52	GPIO46	CE_BQ	Power-2	Out	IE, WPD	Internal WPD	Strap (boot)		C14

Source: [8]; reset states and flags from [2]; strapping from [3].

ID	Comment	Source
C1	No boot access. Add a 10 kΩ pull-up to +3V3_DIG and a boot button to GND on GPIO0, and a reset button to GND on CHIP_PU. Download mode is entered by holding GPIO0 low while CHIP_PU rises.	[2], [3]
C2	Pull-down assignment of R12 to R14 (RESETB, GPIO1, GPIO2) read from the DAC sheet; confirm.	[8]
C3	GPIO3 strap selects the JTAG source only when the eFuse enabling it is burned; floating by default.	[3]

ID	Comment	Source
C4	Two 10 kΩ pull-up pairs on the I2C bus (DAC sheet R10 / R11, charger sheet): 5 kΩ effective. Acceptable at 100 kHz; one pair is redundant.	[8]
C5	Encoder A / B (Peripherals Rev 8) need two GPIOs with pull-ups and RC filters and may wake the player, so they need RTC-capable pins. GPIO15 / GPIO16 are the only free RTC GPIOs.	[9], [3]
C6	Pin budget: 7 free pins against SD, encoder A / B and backlight PWM. Proposed: 1-bit SDMMC on GPIO47 (CLK), GPIO41 (CMD), GPIO42 (D0), card detect on GPIO26 (valid only without PSRAM). 4-bit SDMMC needs D1 to D3: free GPIO1_DAC, GPIO2_DAC and ADDR_DAC (hardwire the address with a resistor). Pull-ups on CMD and data lines per the guideline.	[2], [8]
C7	Backlight control output required by Peripherals Rev 8; driver open there. GPIO47 / GPIO48 sit in the VDD_SPI domain and are 3.3 V only while GPIO45 = 0 and the flash is 3.3 V.	[2], [9]
C8	Power-up glitch pulses the LCD reset; harmless because firmware resets the panel at start-up.	[2]
C9	Delete the wire stubs left on pins 32 to 34, then connect the flash. VDD_SPI (pin 29) becomes the flash supply with decoupling at the pin. Series R and shunt C on SPICLK per the SPI EMC note.	[2], [8]
C10	Floats at reset (R203 / C213 have no DC path). Add a 1 MΩ pull-down per the LTC3440 datasheet; on the GPIO side of R203 it forms no divider when the GPIO drives high.	[10], [8]
C11	The ROM drives U0TXD during boot. A press while TX is low shorts the driver to +3V3 through the switch; a 1 kΩ series resistor limits this to 3.3 mA. Disable the UART0 console in firmware. GPIO43 / GPIO44 are not RTC-capable: UP / DOWN cannot wake the player.	[2], [3]
C12	Schematic buttons: 10 kΩ pull-down, switch to +3V3 (active high). Peripherals Rev 8 specifies pull-up, switch to GND. Align the two.	[8], [9]
C13	Keep the pull-down. A pull-up (usual for CS) makes GPIO45 = 1 at reset and selects 1.8 V on VDD_SPI.	[2]
C14	CE low at reset keeps charging enabled. GPIO46 = 1 with GPIO0 = 0 is an invalid strap combination.	[3]

4. External Components

Ref	AL-PN	Part	Notes	Qty
U		Espressif ESP32-S3	56-QFN, 7 × 7 mm	1
Y	#1-118	ECS Inc. ECS-400-10-37B2-CKY-TR	40 MHz crystal, C _L 10 pF, ±10 ppm tolerance, ±10 ppm stability, ESR 40 Ω, 4-SMD. DigiKey 50-ECS-400-10-37B2-CKY-TRCT-ND. [4], [5]	1
C (XTAL)	#1-122	Murata GJM1555C1H120FB01D	12 pF, ±1 %, 50 V, COG, 0402. Crystal load capacitors, one per crystal pin (C1 / C4 in the guideline figure). DigiKey 490-11190-1-ND. [6]	2
L (XTAL_P)	#1-120	TDK MHQ1005075HA24N HTD25	24 nH, ±3 %, 250 mA, 700 mΩ max, 0402. Series component on XTAL_P; 24 nH is the guideline starting value. DigiKey 445-MHQ1005075HA24NHTD25CT-ND. [2], [7]	1
L (VDD3P3)	#1-121	TDK MHQ1005075HA2N0 BTD25	2 nH, 1 A, 50 mΩ max, 0402. LC filter inductor on the VDD3P3 rail (pins 2 and 3); guideline asks for ≥ 500 mA rating. DigiKey 445-MHQ1005075HA2N0BTD25CT-ND. [2], [7]	1

5. Equations and Results

Crystal load capacitors (C1 = C4)

$$C_L = \frac{C_1 C_4}{C_1 + C_4} + C_{\text{stray}} \quad (1)$$

$$C_1 = C_4 \Rightarrow C_1 = C_4 = 2(C_L - C_{\text{stray}}) \quad (2)$$

$$C_{L, \text{eff}} = \frac{C_1}{2} + C_{\text{stray}} \quad (3)$$

Source: equation (1) and C1 / C4 naming per [2], section External Crystal Clock Source. Equations (2) and (3) follow from (1) with C1 = C4.

Parameter	Value	Source
C _L (crystal load capacitance)	10 pF	[4], [5]
C _{stray} (PCB stray capacitance)	3 to 5 pF (assumed; the guideline gives no value)	Estimate
C1 = C4 required	10 to 14 pF	Eq. (2)
C1 = C4 selected	12 pF (#1-122), centre of the required range	Eq. (2)
C _{L,eff} with 12 pF	9 to 11 pF (C _L = 10 pF ± 1 pF)	Eq. (3)
Tuning	Set after assembly: TX tone test, measure 2.4 GHz frequency offset, trim to within ±10 ppm. Positive offset: increase load capacitance. Negative offset: decrease it.	[2]

6. Guideline Notes

Topic	Note	Source
SPI (EMC)	Place a series resistor or ferrite bead plus a capacitor to ground on the SPI_CLK trace. Where space allows, add the same series R and shunt C on the other SPI traces. Keep the RC / LC parts close to the chip pins.	[2], SPI
XTAL_P	Series component on XTAL_P, 24 nH initial value, adjusted after an overall test, to limit crystal harmonics reaching the RF path.	[2], Clock Source
VDD3P3	LC filter on the VDD3P3 rail to suppress high-frequency harmonics; inductor rated 500 mA or more. 10 µF on the rail recommended because TX bursts can collapse it.	[2], Analog Power Supply
Crystal	Crystal amplitude recommended above 500 mV. Frequency deviation beyond ±10 ppm degrades RF performance.	[2], Clock Source
CHIP_PU	RC delay recommended, typically R = 10 kΩ and C = 1 µF; must not float.	[2], Power-up and Reset
Strapping	GPIO0, GPIO3, GPIO45 and GPIO46 are strapping pins. GPIO0 and GPIO46 set boot mode; GPIO45 sets VDD_SPI voltage (0: 3.3 V, 1: 1.8 V) with default eFuses. GPIO46 = 1 with GPIO0 = 0 is an invalid combination.	[2], [3]

7. Open Problems

#	Problem
1	External SPI flash selection: 3.3 V, quad SPI (octal would take GPIO33 to GPIO37), 8 MB or more, Espressif-validated model, WSON / USON package. VDD_SPI to flash VCC with decoupling.
2	Memory: base chip without PSRAM, or add an external PSRAM (uses SPICS1, which is proposed for SD card detect).
3	Boot access: GPIO0 pull-up and boot button; reset button on CHIP_PU (Section 3.3, C1).
4	SHDN_BB_3.9 pull-down, 1 M Ω (C10).
5	DOWN on U0TXD: 1 k Ω series resistor (C11).
6	GPIO budget: SD bus width, encoder A / B and backlight PWM against 7 free pins (C5 to C7).
7	microSD card and card socket selection.
8	Button polarity differs between the schematic and Peripherals Rev 8 (C12).
9	Wire stubs on flash pins 32 to 34 (C9).
10	Supply decoupling per pin group, per the guidelines.

Source: [2], [3], [8], [9].

8. Revision Log

Rev	Date	Change
0	2026-09-29	Initial issue: ESP32-S3 base chip, 40 MHz crystal, design references, schematic review items.
1	2026-09-29	Crystal changed to ECS-400-10-37B2-CKY-TR (10 pF). Load capacitors, XTAL_P series inductor and VDD3P3 filter inductor added. Pin default configuration and power-up glitch tables added. Guideline notes added (SPI EMC). Guideline source moved to the online ESP32-S3 Hardware Design Guidelines. Open problems 1, 8, 9 updated; 11 and 12 added. All entries cited.
2	2026-09-29	Crystal load capacitors changed to 12 pF GJM1555C1H120FB01D (#1-122); $C_{L,eff}$ 9 to 11 pF recorded. Open problem 1 (load capacitor value) closed; open problems renumbered.
3	2026-09-29	GPIO assignment table (Section 3.3) from the current schematic sheets, with recommended changes and comments. Closed: LCD off flash pins, CHIP_PU RC, VDDA supply, I2C crossing, strapping (EN1_BQ removed, CE_BQ on GPIO46, CS_LCD pull-down on GPIO45), power-up glitch review. Added: boot access, SHDN_BB_3.9 pull-down, U0TXD series resistor, GPIO budget, button polarity, flash-pin stubs.

9. Sources

1. [ESP32-S3 Hardware Design Guidelines \(Espressif\)](#)
2. [ESP32-S3 Hardware Design Guidelines, Schematic Checklist \(Espressif\)](#)
3. [ESP32-S3 Series Datasheet \(Espressif\)](#)
4. [ECS-400-10-37B2-CKY-TR Product Page \(DigiKey\)](#)
5. [ECS-400-10-37B2-CKY-TR Product Page \(ECS Inc.\)](#)
6. [GJM1555C1H120FB01D Product Page \(DigiKey\)](#)
7. [MHQ1005075HA Series Inductors \(TDK, via DigiKey\)](#)
8. DAP schematic sheets DAP-MCU-3, DAP-Audio-1, DAP-Power-2, DAP-Peripherals-4 (author, 2026-09-29)
9. Peripherals: Working Document, Rev 8 (project file)
10. LTC3440 datasheet, SHDN/SS pull-down value (as reported by the author)